

What If the Universe Is NOT Expanding?

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Video Original English: <https://www.youtube.com/watch?v=V4HxyS0M2UM>

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Video Version

The philosopher Thomas Kuhn argued that science is "a series of peaceful interludes punctuated by intellectually violent revolutions" in which "one conceptual world view is replaced by another." (*The Structure of Scientific Revolutions*, 1962) The longest and most tranquil periods of "normal science" are guided by a dominant paradigm. Astrophysics has been in such a peaceful interlude for decades with its belief that the universe is expanding. The expanding universe theory is a paradigm that guides how astrophysicists think about the universe.

As Kuhn further stated, eventually, phenomena will emerge that do not conform to the paradigm. Kuhn describes the process by which science then proceeds. Initially, it dismisses such dissenting data as "anomalies" or minor difficulties that science will eventually resolve and fit into the paradigm. Science resists the idea that the prevailing theory and paradigm are false, and resists alternative theories to explain the anomalies. As anomalous data build up, normal science goes into crisis. Eventually, a new paradigm that better explains the data will emerge and gain a following of scientists who will overthrow the established theory, and a scientific revolution will have occurred. (*How We Are and How We Got Here*, 2022)

The Paradigm

The dominant paradigm in astrophysics is that the universe began in a "big bang," probably from a singularity, and that the universe, the very fabric of space, has been expanding ever since. The expanding universe theory was developed to try to explain observations that many galaxies exhibit a redshift in their spectra, which scientists equated understandably with the well-established phenomenon of the Doppler effect. They assumed, therefore, that a galaxy showing a spectrum with redshift is moving away from us, the greater the redshift, the greater the velocity. (They also assumed that greater redshift means greater distance, a circular reasoning which we will set aside for now.) The fact that so many galaxies appear to be redshifted meant, scientists assumed, that the entire universe is expanding, with galaxies moving away from each other.

The expanding universe theory has the advantage of internal consistency. The rational, mathematical beauty of a theory is seductive to scientists. Fair enough, but for a theory to be a useful paradigm for describing reality, it must also exhibit external consistency — its assumptions and predictions must match observable phenomena. This is where the expanding universe theory falls short; it is not consistent with everything that is the case in the universe.

Anomalous data are pressing ever harder against the dominant paradigm of astrophysics. Just yesterday, news came out of yet another phenomenon that does not conform to the dominant paradigm: very old galaxies that should not exist, in fact, do exist. Said a co-author of the study revealing the new fact: "We expected to find only tiny, young, baby galaxies at this point in time, but we discovered galaxies as mature as our own in what was previously understood to be the dawn of the universe."

I am not surprised. This is yet another big clue that the dominant paradigm about the universe expanding from a "big bang" is incorrect. The observations in the study mentioned are just one sliver in a growing stream of data showing this. Astrophysics has many complexities, but one prediction of the expanding universe theory can be easily checked by observation, and thus the theory can be easily confirmed or refuted.

The Paradigm's Prediction

Cosmologists tell us that the universe is expanding. However, if the universe had been expanding for 12+ billion years, then we would see certain easily noticeable evidence. The expanding universe theory predicts that as space expands, the matter within space becomes less densely distributed and galaxies get ever further apart from each other.

Introductory astronomy courses use the analogy in the graph to explain the theory.

The theory's prediction as explained in elementary astronomy courses.

The claim is that each raisin is moving ever further from all the others. Simple enough to understand. Logically, this means that as the bread continues to grow, over time, the raisins will be ever further from all the other raisins. Therefore, as time passes, the density of raisins within the bread will decrease as the bread expands because the raisins will be more sparsely spread out. We see this phenomenon in the rising bread dough.

But do we see this in the galaxies? The expanding universe theory predicts that as time passes, the density of galaxies within the universe will decrease as the universe expands, moving galaxies ever further from each other, spreading them more thinly across space. The universe has been around for billions of years. It is not as simple as observing the bread dough rising, but we can indeed look at galactic density both by space and by time to confirm or refute the expanding universe theory.

Looking at the Evidence

A light-year is both a measure of time and distance. If an object is one light-year away from us, then light takes one year to reach us. This means that the light we see is what the object was like a year ago. So when astronomers talk about galaxies billions of light-years away, they also mean that their observations are of those objects as they were and *where* they were billions of years ago.

So if we have a distance scale like this of billions of light-years:

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1. . . 1 . . 2 . . 3 . . 4 . . 5 . . 6 . . 7 . . 8 . . 9 . . 10

The numbers indicate both distance *and* time. A galaxy that is 8 billion light-years away is observed by us where it was 8 billion years ago, its light only having reached us after traveling 8 billion years. Same with a galaxy 4 billion light-years away, 2 billion light-years away, and so on.

Observations of galaxies show that when we plot each galaxy observed by its distance from us, we get a uniform distribution. If we compare an incredibly huge section of space centered on a point 2 billion light-years away with a same incredibly huge section at 8 billion light-years away, we see approximately the same density of galaxies at 2 billion light-years away from us as at 2, 3, 4, and so on billion light-years away.

Because distance also equates to time, this means that the density of galaxies is also approximately the same 2, 3, 4, and so on billion *years* ago, so the density of galaxies is approximately the same in all eras. When we observe galaxies at 8 billion light-years away, we are seeing them as they were 8 billion years ago. This allows us to look back in time and very easily check the expanding universe hypothesis.

Going back to our raisin in the bread dough analogy, if we measured the density of raisins per cubic centimeter in the dough at time zero, then came back 1, 2, and 3 hours later and observed that the density of raisins per cubic centimeter in the dough had not changed, we would have to conclude that the bread dough is not rising. Any baker would be foolish to think otherwise. An analogy that was meant to be an elementary example of a cosmological theory ends up becoming a proof of concept that the universe is not expanding.

Reiterating What Should Be Obvious

This is a difficult concept to grasp, so I will reiterate it in different words. (I was shocked at how many astrophysicists cannot grasp these concepts, so please, forgive the repetition.)

The expanding universe theory says that as space expands, over time galaxies spread out and get further from each other and become less densely packed in space. The theory predicts that we would see a lower density of galaxies at 2 billion light-years

away than at 8 billion light-years away. If the universe had been expanding all this time, then galaxies at 8 billion light-years away would be, on average, closer to each other than galaxies at a closer distance. It is simple to measure this and measurements of galactic distribution have been made.

All surveys show that there is no significant change in the relative density of the galactic distribution across the universe, which means no difference over space or time. There is "clumping" in the universe where there are points of greater density of matter. However, the level of this "clumping" is the same at all distances — the density of galaxies in regions eight billion light-years away is not different from the density of galaxies in regions two billion light-years away. Pick any two distances and the same is true. Thus, since distance in space equates to distance in time, the observations confirm that the density of galaxies 8 billion years ago is not different from the density of galaxies 2 billion years ago. Therefore, we must conclude that the expansion theory is contradicted by the evidence.

The data indicate that the density of galaxies in the universe has not decreased over time. Galaxies are not further from each other in more recent times. This defeats the expanding universe theory. The paradigm is wrong, but astrophysicists are still resisting the data showing that the prevailing theory and paradigm are false, and they resist alternative theories to explain the anomalies.

Time for a Revolution or At Least to Move Forward With a New Paradigm

I have no doubt that the universe has been around for billions of years, but the expanding universe hypothesis is clearly false. So why do scientists continue to insist that the universe is expanding? It is classic "group think" — a common malady in all areas of human society. We see it in culture, religion, politics, and even in science. Kuhn was correct. Once an idea becomes popular, it creates an inertia that makes it difficult to sweep away, even in the face of evidence. Beliefs, whether scientific or not, are created and maintained because they serve a useful purpose; but that purpose is not always because it best explains the available evidence. It is all too easy for people to ignore inconvenient truths if denying them serves a purpose.

Sometimes, even brilliant minds are lazy and don't ask questions. This is especially the case when brilliant minds get together and confirmation bias takes over. Science does a good job of correcting itself, even if it takes some time. Kuhn was correct that science is, at first, resistant to evidence that is contrary to its paradigm and contradicts its theories. Scientists are human after all, all too human in their psychological and rational frailties.

Astrophysicists need to create a new paradigm, a new theory that explains the observed redshifts in galaxies. The assumption that this can be explained by the expanding universe is not compatible with other observations. What that theory

should be, I cannot hazard a guess, but that the current theory is wrong is easily seen, though perhaps more easily seen by those on the outside looking in.

References:

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